WEEK 4

PROGRAM 1

```
#include <iostream>
using namespace std;
int main() {
    int n, key;
    bool status = false;
    cout << "Enter n value and key value: ";
    cin >> n >> key;
    int arr[n];
    int low = 0, high = n - 1;
    cout << "Enter all array elements in sorted order: ";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
    while (low <= high) {
        int mid = (low + high) / 2;
        if (arr[mid] == key) {
            cout << key << " element found at " << mid << " location";
            status = true;
            break;
```

```
        }
        else if (arr[mid] < key) {
            low = mid + 1;
        }
        else {
            high = mid - 1;
        }
    }
    if (!status) {
        cout << "Element not found in the array";
    }
    return 0;
}
```

PROGRAM 2

```
#include <iostream>
using namespace std;
int main() {
int n;
cout << "Enter number of elements: ";
cin >> n;
int arr[n];
```

```
cout << "Enter elements:\n";
for(int i = 0; i < n; i++) {
cin >> arr[i];
}
for(int i = 1; i < n; i++) {
for(int j = i; j > 0; j--) {
if(arr[j] < arr[j - 1]) {

int temp = arr[j];
arr[j] = arr[j - 1];
arr[j - 1] = temp;
}
}
}
cout << "Sorted array:\n";
for(int i = 0; i < n; i++) {
cout << arr[i] << " ";
}
return 0;
}
```

PROGRAM 3

```
#include <iostream>

using namespace std;

int main() {

int n;

cout << "Enter number of elements: ";

cin >> n;

int arr[n];

cout << "Enter elements:\n";

for(int i = 0; i < n; i++) {

cin >> arr[i];

}

for(int i = 0; i < n - 1; i++) {

for(int j = 0; j < n - i - 1; j++) {

if(arr[j] > arr[j + 1]) {

int temp = arr[j];

arr[j] = arr[j + 1];

arr[j + 1] = temp;

}

}

}
```

```
cout << "Sorted array:\n";
for(int i = 0; i < n; i++) {
cout << arr[i] << " ";
}
return 0;
}
```